

Divide and conquer algorithms and Master theorem II.

1. Give a recursion. Then solve the recursion by Master theorem.
 - (a) Number of inversions in an array $A[1..n]$. (A pair $1 \leq i < j \leq n$ is an inversion if $A[i] > A[j]$.)
 - (b) Is there a majority element in an array (i.e., an element with more than $\frac{n}{2}$ occurrences)? If there is one, determine it.
2. Solve the recursion for merge sort without Master Theorem.
3. Give a recursion for the towers of Hanoi and solve it.
4. Assume, that we have n pairs of screws (male and female screws) of different sizes are stored in 2 boxes, one is the box of male screws and the other one is the box of the female screws. Give a randomized divide and conquer algorithm to find the pairs. (We assume that the chance to pick one specific screw from the 2 boxes is equal for all the screws.) Show that the given algorithm is efficient, i.e. the expected number of comparisons can be upper bounded by a small degree polynomial.
5. Determine the closest pair among n points on the Euclidean plane by a divide and conquer algorithm. (Points are given by their coordinates.) Solve the recursion by Master theorem.
6. Find the convex hull of n points on the Euclidean plane by a divide and conquer algorithm. Solve the recursion by Master theorem.